Investigating Subtype-Specific Ubiquitination Patterns in Prostate Cancer

Aditi Deshpande

Cieslik Lab

ROGEL CANCER CENTER
MICHIGAN MEDICINE

Prostate Cancer Overview

Clinical Landscape:

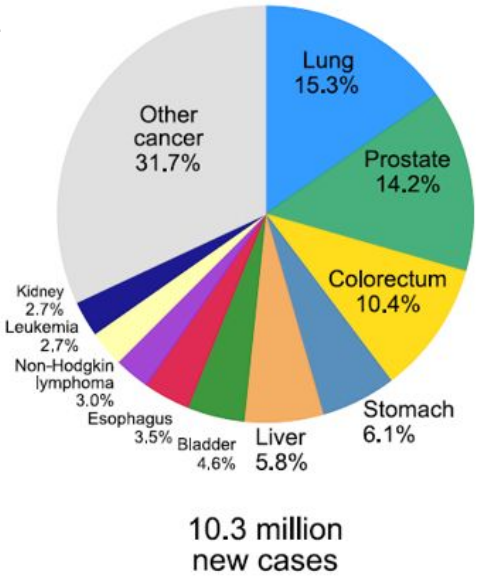
- Second leading cause of cancer death in U.S. men
- Highly heterogeneous: indolent → aggressive disease

Treatment:

- Surgery/radiation (localized); Androgen Deprivation Therapy (systemic)

Global Incidence of PCa in 2024

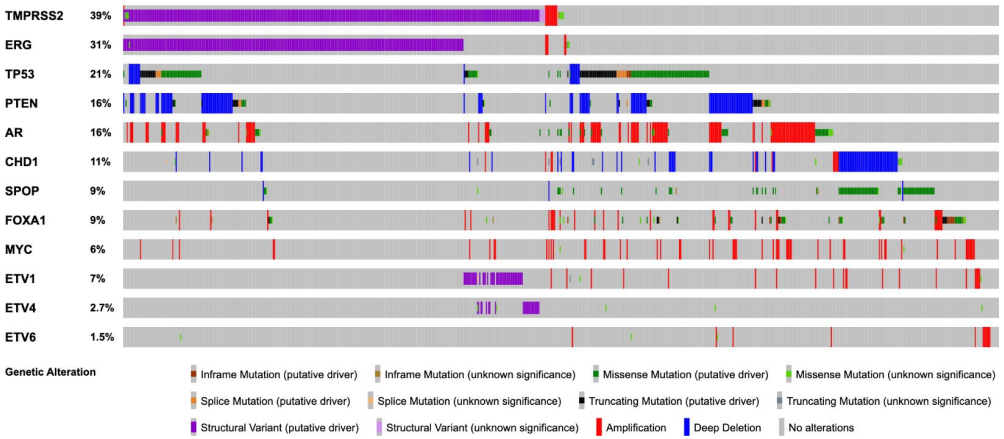
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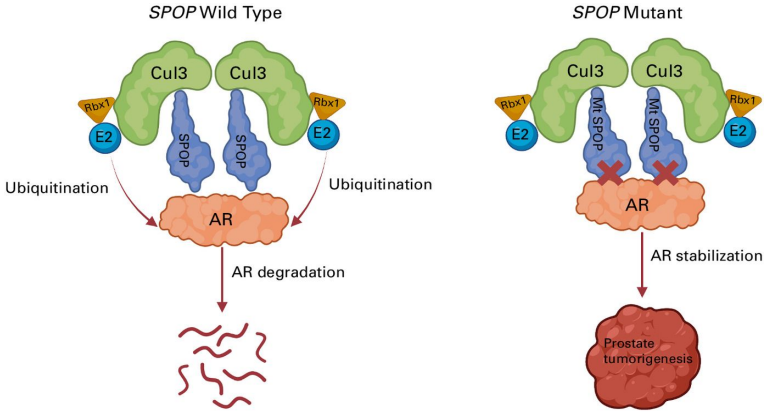
1. American Cancer Society 2026

Prostate Cancer Molecular Overview

- Subtypes: ETS, FOXA1, **SPOP**, Rare
- Driver events are generally **mutually exclusive**
- SPOP functions as a CUL3-based E3 ligase adaptor, directing substrate ubiquitination
  - Targets proteins for ubiquitination → degradation or signaling



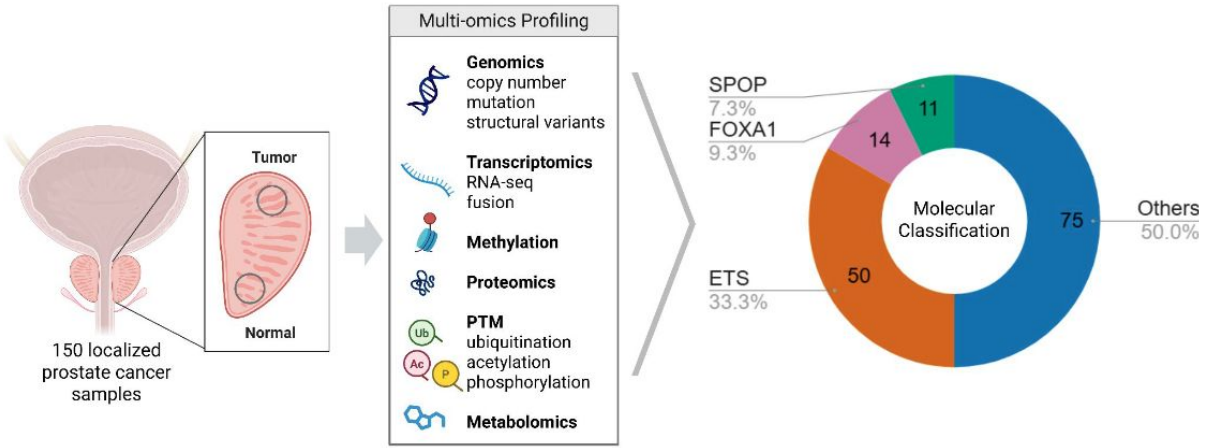
Prostate Adenocarcinoma (MSK/DFCI, Nature Genetics 2018)



Georges Gebrael et al. Differential Impact of SPOP Mutation in Prostate and Endometrial Cancers. JCO Precis Oncol 7, e2300397(2023). DOI:10.1200/PO.23.00397

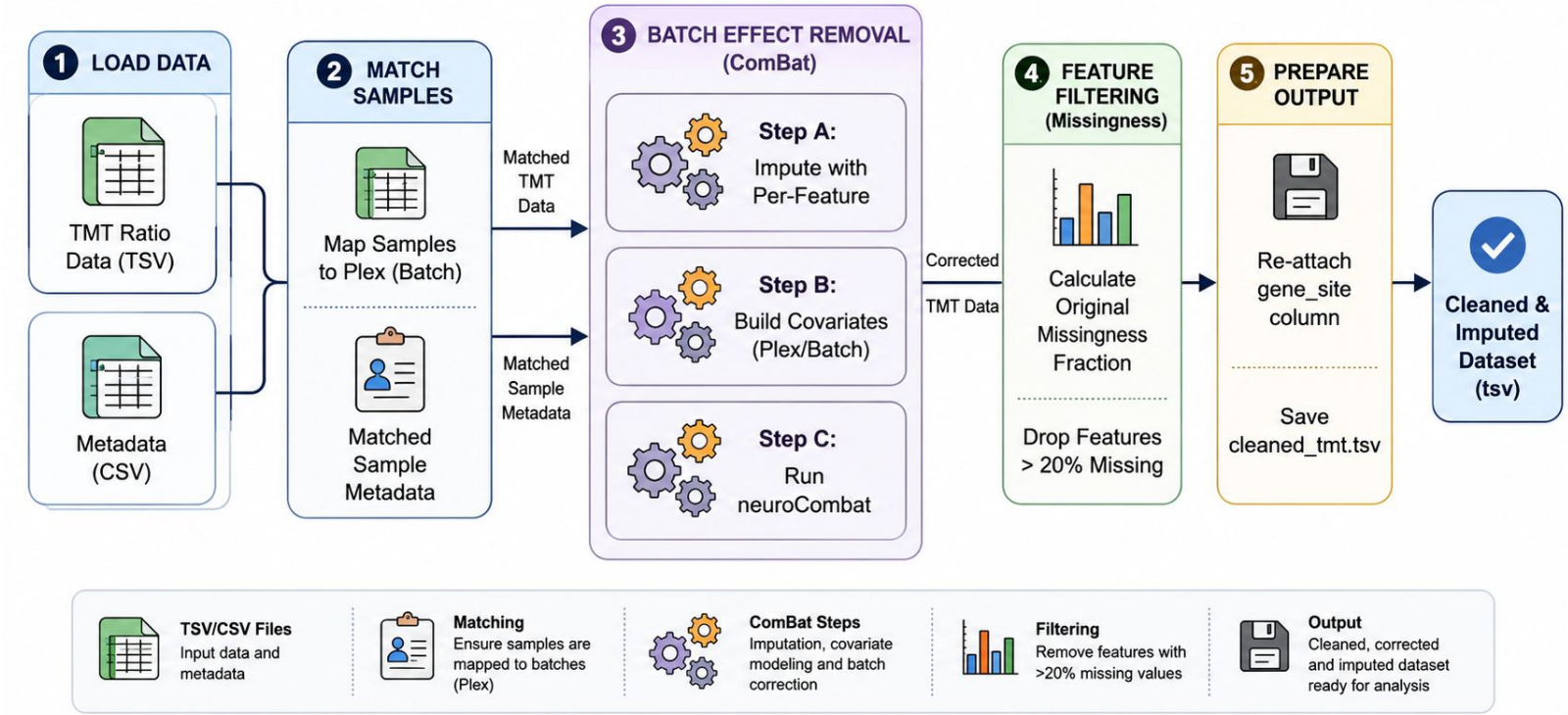
Objective: Ubiquitination Analysis

- Identify subtype-specific ubiquitination patterns across tumors
- Focus on SPOP-mutant tumors
- Integrate multi-omic data



Designed by Mahnoor Gondal using BioRender

TMT Proteomics Data Analysis Pipeline

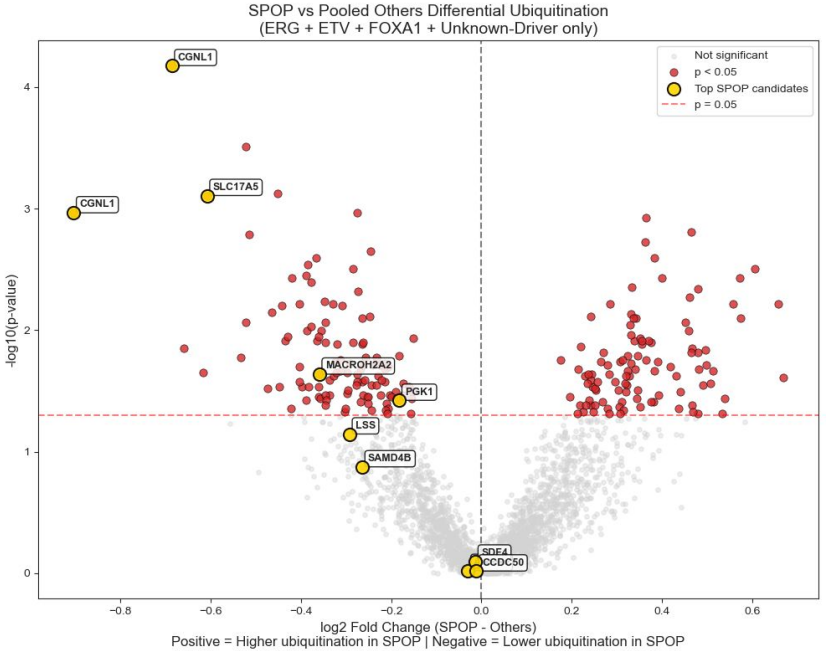


Designed using Gemini AI & ChatGPT

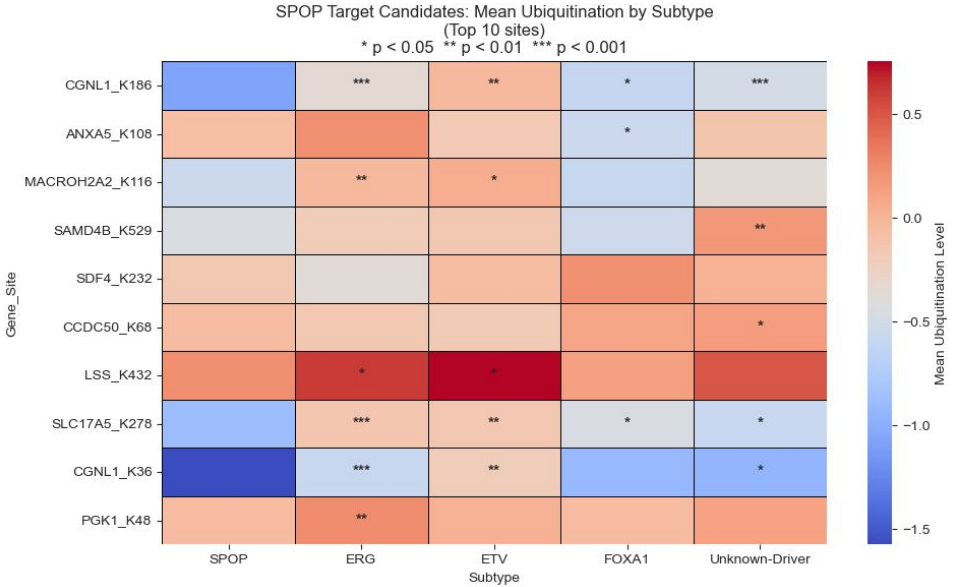
Abundance and Purity corrected data was used.

SPOP-focused Analysis

Goal: Find proteins whose ubiquitination drops specifically in SPOP tumors

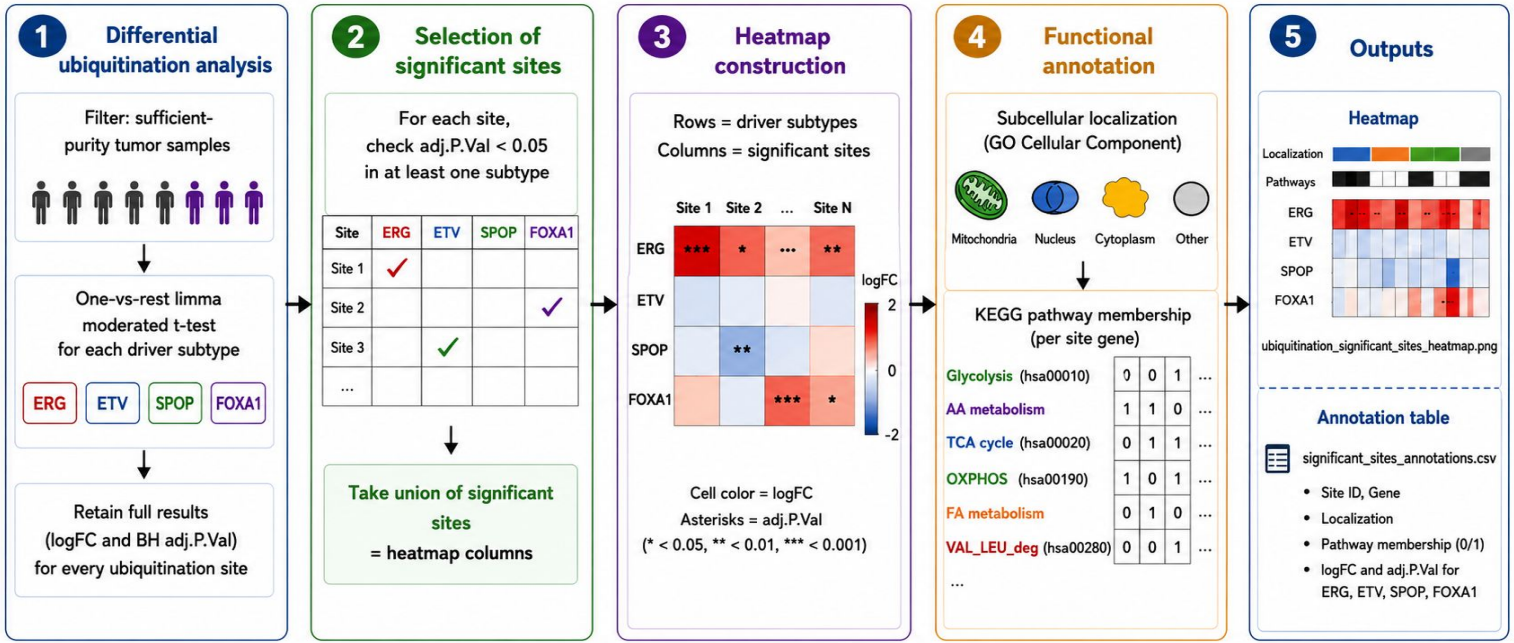


Pooled SPOP-vs-rest



Pairwise Consistency Anova vs. Each Subtype

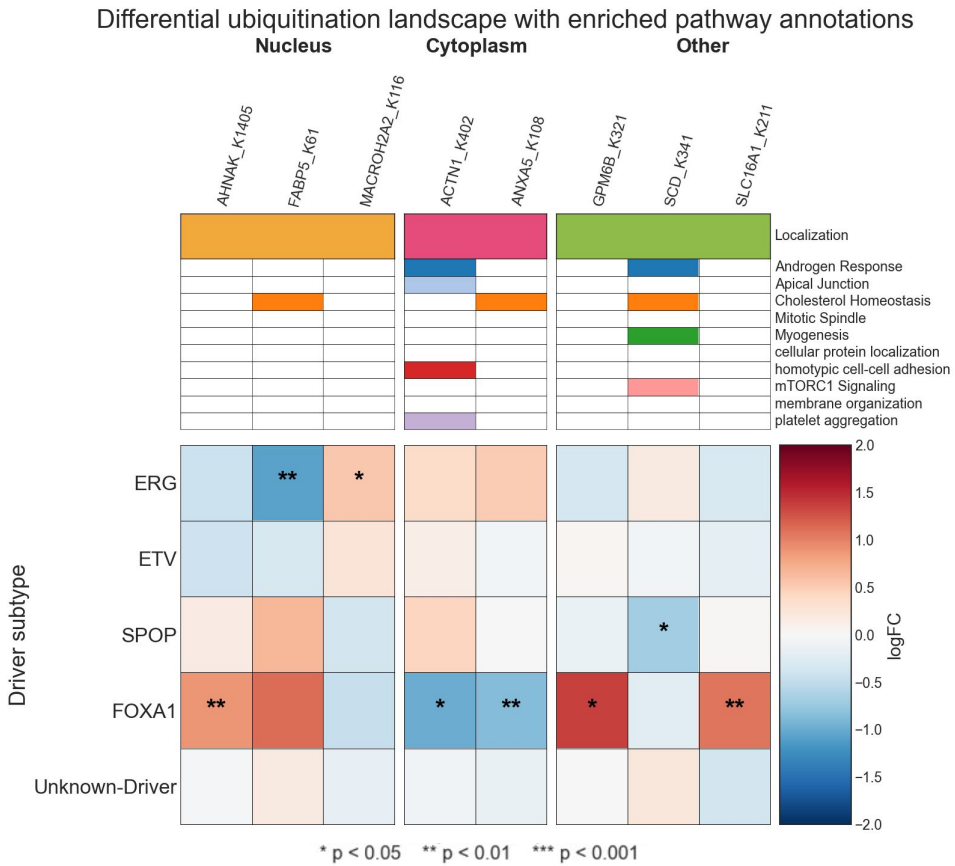
Cellular Compartment and Pathway Focused analysis



Designed using ChatGPT

Abundance and Purity corrected data was used.

Differential Ubiquitination Site Landscape by Compartment and Pathway



- **8 differential ubiquitination sites** identified (BH-adjusted $P < 0.05$).
- Other category consists of membrane-associated proteins

Future Directions

- Validate findings with DIA data
- Literature review

Challenges and Technical Considerations

- Reconciling DIA vs TMT discrepancies for cross-omics validation
- Consistent PTM-aware imputation incorporating tumor purity and abundance among the omic-layers

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